

Abstract. We compare two string kernels for SMS spam detection, a character spectrum (k -mer) kernel and a Levenshtein edit-distance kernel, against a word TF-IDF baseline, all trained with the same soft-margin kernel SVM. On clean text the word baseline and the spectrum kernel are statistically indistinguishable ($\sim 97\%$ accuracy, ROC-AUC ≈ 0.99), while the edit-distance kernel trails at the default threshold. We show that this kernel is not positive semi-definite, that the naive diagonal PSD repair over-regularizes into a degenerate classifier, and that a kernel-PCA projection restores it. Our central result is adversarial: when spam is obfuscated (leetspeak, intra-word spacing) the clean-data ranking inverts. Character n -grams collapse, word features degrade, and the edit-distance kernel becomes the most robust representation. A convex mixture of the two kernels hedges the tradeoff: $\alpha = 0.25$ keeps the full robustness at near-clean accuracy. The best kernel depends on the threat model, not on clean-data accuracy.

1 Introduction

Spam detection on short, noisy text is a natural fit for kernels that compare strings by character structure rather than by a fixed bag-of-words vocabulary. We study two such kernels. The **spectrum kernel** counts shared length- k substrings; the **Levenshtein kernel** turns edit distance into a similarity. We benchmark both against a standard word TF-IDF + linear-SVM baseline, with the string-kernel models trained through the same `courselib BinaryKernelSVM`, so that among them only the representation varies, not the optimizer. As a second model from the course we add a multilayer perceptron (week 10) on the same TF-IDF features as the baseline, isolating the model family itself.

Beyond the required comparison, we contribute two pieces of analysis. First, the edit-distance kernel is indefinite, and we show that the obvious fix breaks the classifier while a kernel-PCA projection repairs it, a distinction that ROC analysis makes precise. Second, and centrally, we test each representation under adversarial obfuscation, the evasion setting that motivates character-level methods in the first place. This turns an otherwise inconclusive clean-data comparison into a clear, practically relevant result.

2 Data and methods

Data. The SMS Spam Collection contains 5,574 messages, $\sim 13\%$ labelled spam. Preprocessing lowercases and collapses whitespace but keeps digits and punctuation, which carry character-level spam signal. We draw a stratified subsample of 1000, split 75/25, and report every number as mean $\pm$ std over 5 seeds.

Kernels. For strings s, t over alphabet Σ , the spectrum kernel of order k and its cosine normalization are

$$K_k(s, t) = \sum_{u \in \Sigma^k} \phi_u(s) \phi_u(t), \quad \hat{K}_k(s, t) = \frac{K_k(s, t)}{\sqrt{K_k(s, s) K_k(t, t)}},$$

where $\phi_u(s)$ counts occurrences of substring u (the same spectrum kernel used for DNA sequence comparison in week 8, here over the SMS character alphabet). Normalization is essential: spam messages are $\sim 2\times$ longer than ham, so raw counts encode a length bias rather than lexical content. The Levenshtein kernel applies an RBF to the length-normalized edit distance $d \in [0, 1]$,

$$K_\sigma(s, t) = \exp\left(-\frac{d(s, t)^2}{2\sigma^2}\right).$$

We use $k = 3$ and $\sigma = 0.45$, both validated by a sweep (Fig. 1), and soft margin $C = 1$: every kernel is normalized to unit diagonal, so a single C is comparable across kernels. The width σ has a narrow useful regime: below ~ 0.3 the kernel collapses toward the identity, while above ~ 0.7 indefiniteness grows. For the order k , recall is an inverted U peaking at $k = 3$ (precision stays ≈ 1.0 throughout): smaller k clusters spam less tightly, larger k overfits.

The edit-distance kernel is indefinite. Edit distance is not a Euclidean metric, so K_σ is not PSD: at $\sigma = 0.45$ the training Gram has 47/750 negative eigenvalues ($\lambda_{\min} \approx -4.1$), and `cvxopt` requires a PSD matrix. Three repairs exist. (i) A diagonal shift $K + \rho I$ ($\rho = -\lambda_{\min}$) is train/test consistent, but $\rho \approx 4$ dwarfs the ~ 0.2 off-diagonals, leaving a diagonally dominant Gram that over-regularizes the SVM into majority-class prediction. (ii) Eigenvalue clipping alters off-diagonals, making the train and test kernels inconsistent. (iii) We instead project onto the positive eigen-spectrum (kernel PCA): with $K = V\Lambda V^\top$ we keep $\Lambda_+ > 0$, form explicit features $\Phi = V_+ \Lambda_+^{1/2}$, and project test points by the Nyström rule $\Phi_{\text{test}} = K_{\text{test}} V_+ \Lambda_+^{-1/2}$, then train a linear SVM. This is consistent and never inflates the diagonal.

Second model. The required second course model is a multilayer perceptron (week 10) on the same TF-IDF features as the linear baseline: one hidden layer of 64 ReLU units, softmax output, cross-entropy loss, mini-batch gradient descent ($\eta = 1.0$, 300 epochs, batch 64). Sharing the features isolates the change from a margin SVM to a backprop network.

3 Results on clean text

Table 1 reports the comparison. The word baseline and the spectrum kernel are **statistically indistinguishable** (97.0 vs. 97.5% accuracy; AUC 0.992 vs. 0.986). This is expected: the spectrum kernel is a linear kernel in k -mer count space, so it is a linear text classifier of the same family as the baseline. They differ only in operating point (spectrum favours precision ~ 1.0 , the baseline favours recall), and the ROC curves (Fig. 2) confirm this is a threshold effect, not a capability gap: the curves are nearly coincident.

The two Levenshtein rows tell the methodological story. The direct diagonal-shift model is degenerate

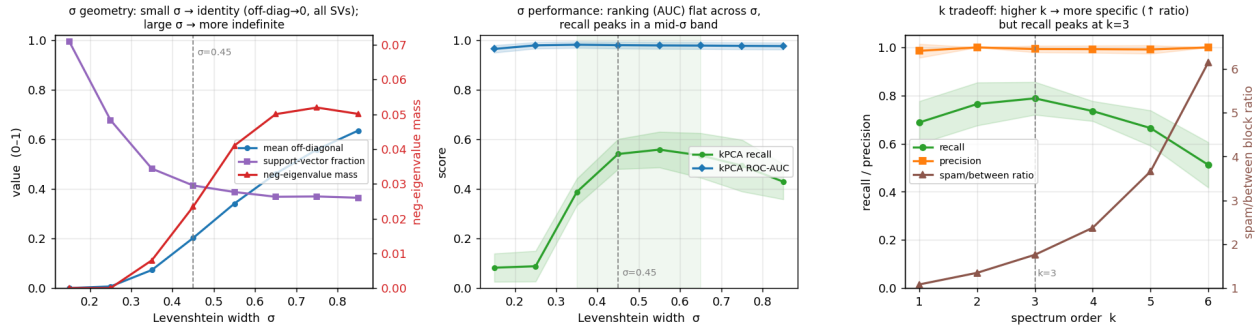


Figure 1. Hyperparameter sensitivity (mean over 5 seeds, band = ± 1 std). **Left:** as $\sigma \rightarrow 0$ the Levenshtein kernel collapses toward the identity; larger σ raises the negative-eigenvalue mass. **Middle:** the kPCA classifier's ranking is flat across σ ; only its default-threshold recall shows a sweet band (shaded), the same ranking-vs-threshold split as Sec. 3. **Right:** larger k is more specific (block ratio rises) but recall peaks at $k = 3$; precision ≈ 1.0 for all k .

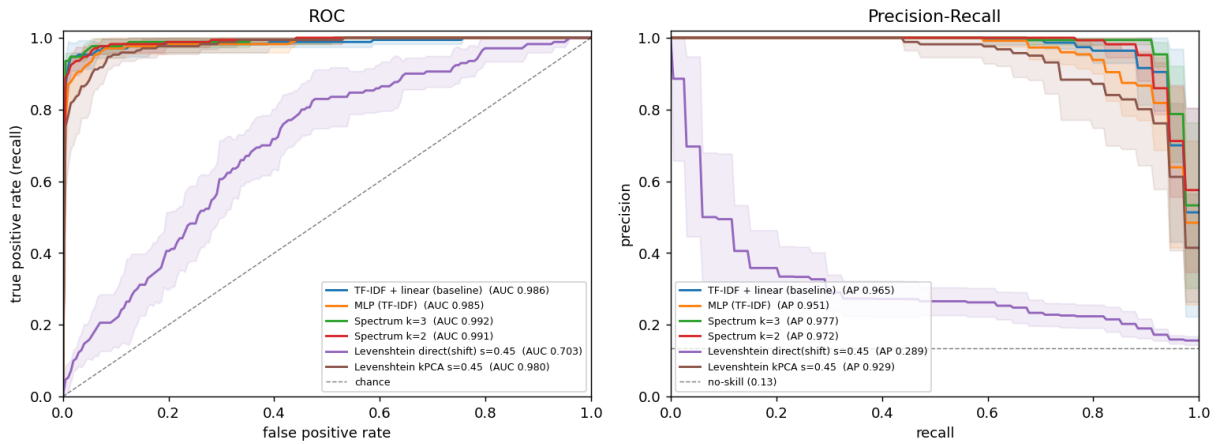


Figure 2. ROC (left) and precision-recall (right), mean over 5 seeds, band = ± 1 std. By threshold-independent ranking the strong models (TF-IDF, the MLP, and both spectrum kernels) cluster at AUC ≈ 0.98 – 0.99 , so the recall gaps in Table 1 are threshold artifacts. The direct shift kernel ranks above chance (AUC 0.70) but collapses in precision (AP 0.29); its failure is threshold placement, not lost signal, which the kPCA projection restores (AUC 0.98).



Figure 3. Training kernel matrices, rows/cols sorted by class; the white box outlines the spam-spam block. The spectrum kernel makes spam 1.8 $\times$ more self-similar than between-class pairs (bright block); the Levenshtein kernel leaves all pairs near 0.2 (1.1 $\times$). Within-ham similarity equals between-class similarity for both, so ham is diverse and does not cluster.

(71.8% accuracy, below the 86.4% majority baseline, recall 0.25 with huge variance as it flips between the two constant predictors). Yet its ROC-AUC is 0.70, well above chance, so the kernel does rank spam above ham; only its decision threshold is wrecked by the diagonal dominance. The kernel-PCA projection recovers the same kernel to AUC 0.98 and a stable, high-precision classifier. ROC therefore cleanly separates ranking quality from

threshold placement: the crux of why the naive repair fails and the projection works.

The MLP baseline reaches the same conclusion from the other side. On the identical TF-IDF features it matches the linear SVM in ranking (AUC 0.985 vs. 0.986) yet gives only 0.40 recall at perfect precision. With 13% spam the cross-entropy objective is dominated by ham, so the softmax boundary drifts toward the majority class and the net claims spam only when very confident: a miscalibrated operating point, not a weaker model, exactly as the matched AUC shows (hinge loss with soft margin C balances the classes better, recall 0.84). A second model family thus reproduces the ranking-vs-threshold split independently of the kernel story.

Figure 3 explains the clean-data ordering geometrically. The spectrum kernel isolates a visible spam cluster (1.8 $\times$ denser than between-class similarity); the Levenshtein kernel washes it out (1.1 $\times$). Edit distance between any two short English messages is only ever moderate (they share spaces, vowels, and common bigrams), so the clean signal is weak.

Table 1. Clean-text comparison, mean $\pm$ std over 5 seeds. Majority-class (all-ham) accuracy is 86.4%. † kept to expose the failed PSD repair.

Model	Acc. %	Prec.	Rec.	AUC	#SV
TF-IDF + linear	97.5 $\pm$ 1.0	0.98	0.84	0.986	343
MLP (TF-IDF)	91.8 $\pm$ 0.6	1.00	0.40	0.985	–
Spectrum $k=3$	97.0 $\pm$ 0.9	0.99	0.79	0.992	291
Spectrum $k=2$	96.8 $\pm$ 1.2	1.00	0.76	0.991	189
Levenshtein shift†	71.8 $\pm$ 28.5	0.33	0.25	0.703	419
Levenshtein kPCA	93.7 $\pm$ 0.9	0.99	0.54	0.980	311

4 Robustness to adversarial obfuscation

Clean accuracy is not why string kernels exist; evasion robustness is. We model the realistic threat: spammers obfuscate their messages, legitimate senders do not. We train each representation on clean data, then corrupt only the spam test messages with leetspeak substitution (free $\rightarrow$ fr33) and intra-word spacing, at increasing intensity, and measure spam recall. Because ham is untouched, the false positives are fixed: precision never drops below 0.93 (0.97 for both string kernels), so every change in recall is a genuine detection effect rather than a shift toward predicting spam.

Figure 5 is the central result. The clean-text ordering inverts. Character n -grams, usually assumed robust to obfuscation, collapse the hardest (0.79 $\rightarrow$ 0.12): leetspeak substitutes the very characters the k -grams are built from, and each swap corrupts up to k trigrams. Word TF-IDF degrades more gracefully (0.84 $\rightarrow$ 0.28), losing only the words actually hit. The edit-distance kernel improves (0.54 $\rightarrow$ 0.76): heavily mangled spam is anomalous, far from clean ham in edit distance, and the kernel reads “unlike normal text” as spam.

Key result. The representation that is weakest on clean text (edit distance) is the only one robust to adversarial obfuscation. The clean ranking word $\gtrsim$ char $>$ edit inverts to edit $\gg$ word $>$ char; the lines cross at obfuscation $\approx$ 0.2. The right kernel depends on the threat model.

The reading is that the edit-distance kernel functions as an anomaly detector, and obfuscation increases anomaly. Leetspeak is the sharper half of the attack by construction: substitutions fire four times as often as space insertions, and each swapped character corrupts up to k of the k -mers passing through it, which is exactly the signal the spectrum kernel is built from.

Do we have to choose? No. Interpolating the two kernels, $K_\alpha = \alpha K_{\text{spec}} + (1 - \alpha) K_{\text{edit}}$ (identical kPCA treatment for every α ; the endpoints reproduce Table 1), shows the tradeoff is strongly non-linear in our favour (Fig. 4): clean accuracy saturates by $\alpha \approx 0.5$ while robustness only collapses beyond it. At $\alpha = 0.25$ the mixture keeps essentially all of the edit kernel’s robustness (recall 0.81 ± 0.06 at attack 0.6, at or above pure edit’s 0.76 ± 0.05) while recovering most of the clean gap (accuracy 96.2 ± 1.1 vs. 93.7 pure edit, 97.0 pure spectrum). At mid attack the mixtures even edge out both parents (within ~ 1 std): surviving trigrams and edit anomaly are complementary signals. The honest caveat remains: all of this holds under the “obfuscate spam only” model; an attacker who knew the mechanism would instead make spam look more ham-like, which no fixed representation defends

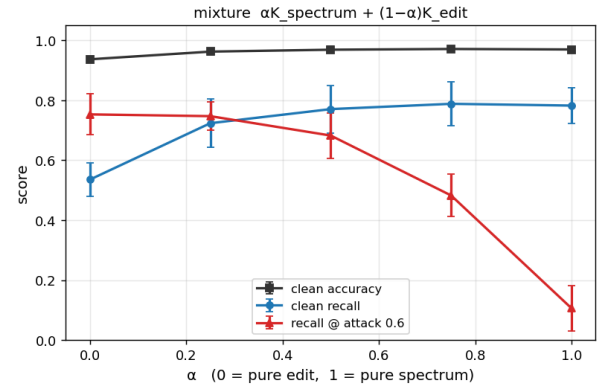


Figure 4. Kernel interpolation (mean $\pm$ 1 std over 5 seeds). Clean accuracy saturates by $\alpha \approx 0.5$; robustness at attack 0.6 holds until $\alpha \approx 0.5$ and then collapses. $\alpha = 0.25$ hedges the threat model almost for free.

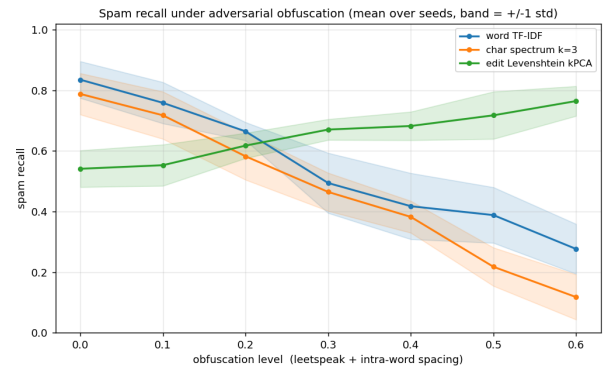


Figure 5. Spam recall vs. obfuscation intensity (mean over 5 seeds, band = ± 1 std). Models trained on clean data; only spam test messages are obfuscated. The clean-data ranking inverts under attack.

against for free.

5 Conclusion

On clean SMS text the choice between word and character n -gram features barely matters: word TF-IDF and the character spectrum kernel are statistically tied, both near 97% accuracy. Our contribution is threefold. Methodologically, we give the indefinite edit-distance kernel a correct treatment (kernel-PCA projection rather than a diagonal shift) and use ROC to separate the kernel’s intact ranking from its broken threshold; the same lens explains the MLP baseline, whose low default-threshold recall masks a ranking on par with the SVM. Empirically, the edit-distance kernel, the weakest on clean data, is the only representation resilient to adversarial obfuscation, while character n -grams, the textbook “robust” choice, are in fact the most fragile to character substitution. Constructively, a convex kernel mixture hedges the threat model almost for free: $\alpha = 0.25$ retains full robustness at near-spectrum clean accuracy. The right representation is decided by the threat model, but you can buy insurance. All results are reproduced by the scripts in the repository (see the README) and persisted to JSON. The study is confined to a 1000-message subsample and non-adaptive obfuscation; scaling to the full corpus and attacks that target the anomaly mechanism directly are natural extensions.

References. Lodhi et al. (2002), *Text Classification Using String Kernels*, JMLR. Ristad & Yianilos (1998), *Learning String Edit Distance*.